

Finding FinCrime when we don't know what to look for

Financial criminals are becoming increasingly sophisticated, and so are the tools needed to catch them. Trust and transparency are key to successful deployment. Models that emphasize explainability and auditability can act as virtual teammates, working in partnership with their human operators.

Artificial intelligence is an essential tool in the fight against financial crime, but how do we learn to trust the AI super sleuth?

Understanding is the key. Fundamentally, AI is a crime-fighting partner, not a black box. Knowing how AI models work to detect FinCrime can help to demystify some common misconceptions and build trust.

Domain knowledge

The starting point for any good model is a basic understanding of FinCrime. To ensure quality outputs, we need to make sure that the model is working with good input data. Domain knowledge is critical to this process.

"Today's approaches to ML and AI are generally domain-agnostic, ignoring domain knowledge that extends far beyond the raw data itself," said the authors of the Argonne National Laboratory paper AI for Science. "Improving our ability to systematically incorporate diverse forms of domain knowledge can impact every aspect of AI."

Anomaly detection is at the core of FinCrime models and relies on the incorporation of such contextual knowledge. This starts with the basic characteristics of financial data, such as transaction size, currency, time, and location, through to more detailed characteristics and statistical relationships.

Machine learning helps to identify anomalies far more efficiently than is otherwise possible. Data scientists decide what features to input into a machine learning model, but they do not decide which features are important to the model. Anomaly detection models can determine the important features on their own.

Knowing that the model has been trained on high-quality data that has been prepared appropriately gives operators the confidence they need to trust its outputs.

Proprietary risk factor and pattern repositories

Because we can use the same risk factors and patterns with multiple financial institutions, the repositories we use grow over time.

We develop these factor repositories with domain experts. One example is Benford's Law, which can be a useful statistical technique to flag questionable financial transactions. To put it simply, Benford's Law says that the numeral 1 will be the leading digit in a genuine data set of numbers 30.1% of the time, while the numeral 2 will be the leading digit 17.6% of the time. Each subsequent numeral through 9 will be the leading digit with decreasing frequency.

This produces a distinctive distribution — it looks like a slide. These consistent distributions

appear in real-world data sets across domains, from stock prices to likes on social media. However, Benford's Law doesn't apply universally. It's important to understand the nature of the data set and the expected distribution of numerals. At Hawk AI, we have observed that many merchant transactions follow Benford's Law, so we added it as a risk factor. Now the model can detect transactions that don't follow these distributions and raise alerts.

Positive feedback loops

AI models learn as they are corrected. As more cases are reviewed by operators, the model gets better at detecting FinCrime. In this sense, AI isn't replacing human operators but working in partnership with them — a virtual teammate.

This is fundamentally different to traditional rules-based screening and transaction monitoring, where human operators repeatedly correct the same mistakes with little or no improvement in the overall result. As well as being fruitless, it can be boring, repetitive work.

With an AI model, the human operators are constantly improving the quality and accuracy of the output. The decisions that operators make are incorporated into the model itself. This doesn't just produce better results for the model, but also helps the human operators to trust the results of AI

and machine learning models. They know that their subject matter expertise played an important role in the results.

Explainability

Financial regulators are increasingly stressing the importance of explainable AI, and rightly so.

Trusting the outputs produced by an AI model requires that users can understand how it makes decisions. Indeed, this is a broad principle of AI in all sectors and not only FinCrime. As AI becomes more complex, rules around transparency and responsible development are a topic of growing significance.

At Hawk AI, our models produce natural language explanations for each risk factor when an anomaly is detected. In its role as a team member, the model supports the investigation by allowing human operators to look into specific transaction and risk factor groupings. The model generates these explanations in plain language that anyone can understand. Operators don't need to be specialists in data science to understand why the AI identified specific activities as suspicious.

Auditability

AI should be auditable as well as explainable, particularly as regulations around the use of AI in finance continue to develop. If a financial institution can't rely on AI solutions to provide results that regulators can audit, the solution isn't much use at all.

Auditability is also a powerful and important component of trust. Not only should a good model explain itself in plain language, but every decision it makes should also be logged and reproducible. This is a core feature of Hawk AI's models. Tracking artifacts show how the model was trained, why it made a certain decision, and why it gave the score it did. If an auditor asks a financial institution these questions, the model will provide the answers.

Model validation

Before any model is deployed, it goes through a validation process, which looks like this:

- Stage 1: Set strict criteria for model deployment.
- Stage 2: Set performance benchmarks for models.
- Stage 3: Compare new model iterations to previous models.

- Stage 4: Make adjustments based on the model's performance.

When FinCrime investigators know how a model was developed and validated, they can rest assured that it will generate reliable results.

Building trust

The effort that goes into developing and validating models creates the foundation for strong and effective partnerships with operators. Together, AI super sleuths and their human teammates can detect anomalies with unmatched speed, accuracy, and transparency.

With criminals now using their own AI, financial institutions of all sizes need to take the threat seriously. Rules-based transaction monitoring systems can't keep pace with the development of modern FinCrime. Catching these criminals requires a sophisticated suite of tools to detect anomalies as they happen. AI models can learn the techniques and spot them in the future, as well as detecting previously unknown risks, all while explaining what they're doing and why. All of this results in AI models that operators can trust to detect financial crime.